Programming Language TASK1

2022186기 이채은

Chapter 3 - Problem Set 3, 4, 7, 8, 13, 14, 16, 19, 22, (2,3,4,5), 23

19, 22 잘 모르겠다!!

3. *에 우선순위 / right associative (+보다)

<assign> → <id> = <expr>
<id> → A | B | C
<expr> → <term> + <expr> | <term>
<term> → <factor> * <term> | <factor>
<factor> → (<expr>) | <id>

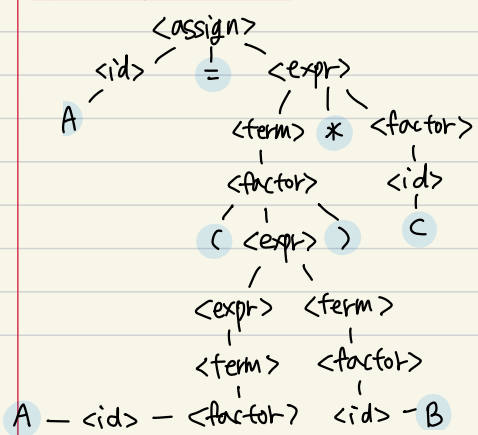
4. ++, -- 단항 연산자 추가

Q1. 전위? 후위?

<assign> → <id> = <expr>
<id> → A | B | C
<expr> → <expr> + <term> | <term>
<term> → <term> * <factor> | <factor>
<factor> → (<expr>) | <id> | <unary>
<unary> → ++<id> | --<id>

7. parse tree & leftmost derivation for each statement

1) A = (A+B) * C

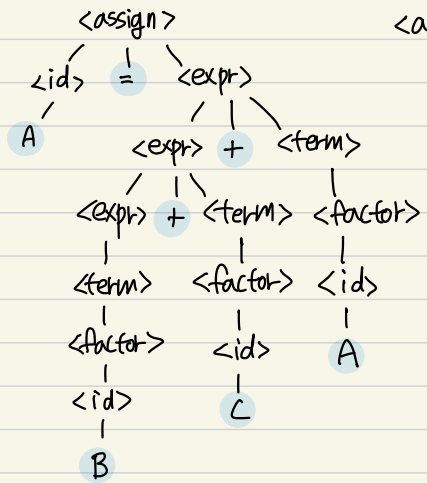


Q2. 구체적으로 다 써야 하는지..?

<assign> → <id> = <expr>
→ A = <expr>
→ A = <term>
→ A = <term> * <factor>
→ A = <factor> * <factor>
→ A = (<expr>) * <factor>
→ A = (<expr> + <term>) * <factor>
→ A = (A + <term>) * <factor>
→ A = (A+B) * <factor>
→ A = (A+B) * C
<factor> → <id>
<term> → <factor> → <id>

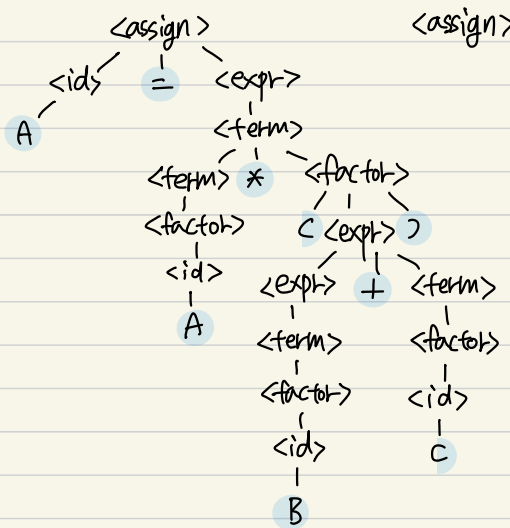
2) A = B+C+A

Q3. Leftmost 가 맞는가?



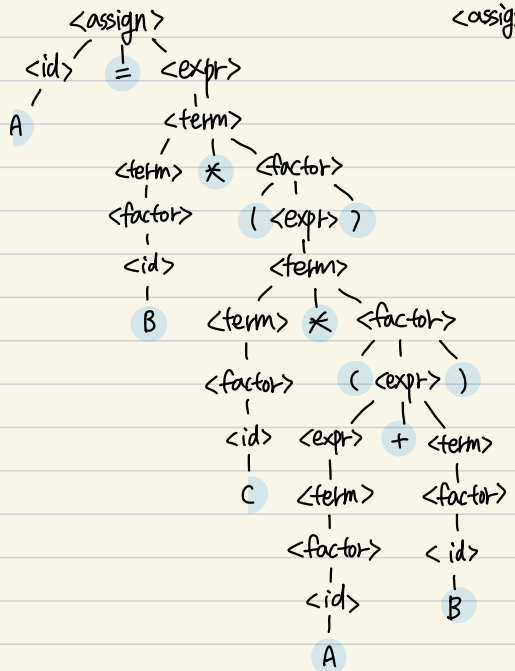
<assign> → <id> = <expr>
→ A = <expr>
→ A = <expr> + <term>
→ A = <expr> + <term> + <term>
→ A = <term> + <term> + <term>
→ A = <factor> 〃
→ A = <id> 〃
→ A = B + 〃
→ A = B + <factor> 〃
→ A = B + <id> 〃
→ A = B + C + 〃
⋮

3) A = A * (B+C)



<assign> → <id> = <expr>
→ A = <expr>
→ A = <term>
→ A = <term> * <factor>
→ A = <factor> * <factor>
→ A = <id> * <factor>
→ A = A * <factor>
→ A = A * (<expr>)
→ A = A * (<expr> + <term>)
→ A = A * (<term> + <term>)
→ A = A * (<factor> + <term>)
→ A = A * (<id> + <term>)
→ A = A * (B + <term>)
→ A = A * (B + <factor>)
→ 〃 <id>)
→ A = A * (B + C)

4) A = B * (C * (A+B))



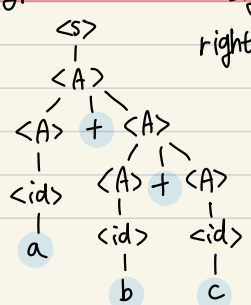
<assign> → <id> = <expr>
→ A = <expr>
→ A = <term>
→ A = <term> * <factor>
→ A = <factor> * <factor>
→ A = <id> * <factor>
→ A = B * <factor>
→ A = B * (<expr>)
→ A = B * (<term>)
→ A = B * (<term> * <factor>)
→ A = B * (<factor> * <factor>)
→ A = B * (<id> * <factor>)
→ A = B * (C * <factor>)
→ A = B * (C * (<expr>))
→ A = B * (C * (<expr> + <term>))
→ A = B * (C * (<term> "
→ " <factor> "
→ " <id> "
→ A = B * (C * (A + <term>))
→ " <factor> "
→ " <id> "
→ A = B * (C * (A+B))

<S> → <A>
<A> → <A> + <A> | <id>
<id> → a | b | c

8. Prove that the grammar is ambiguous

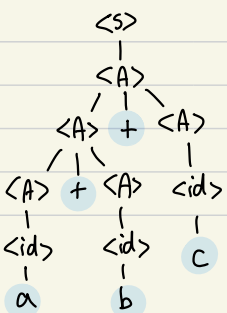
e.g. a+b+c

parse tree ①



right

parse tree ②



left

a+b+c 문장에 대해
leftmost와 rightmost로
parse tree가 2개가 가능
↳ 모호하다..!

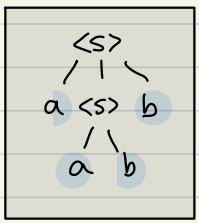
{ a, abb, ba, … (x)
{ ab, aaaabbbb, … (o)

13. a(n) = b(n), n>0 <S> → ab | a<S>b

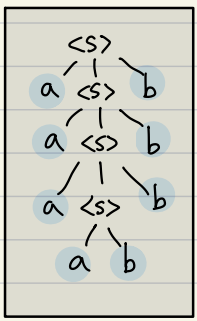
14. Draw the parse trees for aabb and aaaabbbb

Q4. 이렇게가 아닌거..

<S>
a<S>b
aabb …?



aabb



aaaabbbb

16. BNF → EBNF

[ ] : 선택 0 or 1
{ } : 반복 0↑
( ) : 그룹
| : 대안

<assign> → <id> = <expr>
<id> → A | B | C
<expr> → <term> { (+ | *) <term> }
<term> → (<expr>) | <id>

↳ A=B의 경우 0번 반복

유저가 이해하기 어렵다…

19. attribute grammar : language rules 연산자 양쪽에 동일한 데이터 유형 x ?

1. Syntax rule : <assign> → <var> = <expr>
   Semantic rule : <expr>.expected_type ← <var>.actual_type

2. Syntax rule : <expr> → <var>[2] + <var>[3]
   Semantic rule : if (<var>[2].actual_type == int)
   and (<var>[3].actual_type == int)
   then <expr>.actual_type ← int
   else <expr>.actual_type ← real
   end if

Q5. 교재에는 = 안되고, == 써야 되는지?

3. Syntax rule : <expr> → <var>
   Semantic rule : <expr>.actual_type ← <var>.actual_type
   predicate : <expr>.actual_type == <expr>.expected_type

4. Syntax rule : <var> → A | B | C
   Semantic rule : <var>.actual_type ← look-up (<var>.string)

22. denotational semantics mapping functions

2) Java do-while

[[do {S} while (E)]] (δ) =
  let δ' = [[S]] (δ) in
    if [[E]] (δ') is true
    then [[do {S} while (E)]] (δ')
    else δ'

{ loop body : S
  state : δ
  condition : E

3) Java Boolean expressions

[[E]] (δ) = true if E evaluates to true under state δ

[[E]] (δ) = false if E evaluates to false under state δ

{ expression : E
  state : δ

4) Java for

[[for (S1; E; S2) {S3}]] (δ) =
  let δ' = [[S1]] (δ) in
  if [[E]] (δ') is true
  then let δ'' = [[S3]] (δ') in
    [[S2]] (δ'')
    [[for (S1; E; S2) {S3}]] (δ'')
  else δ'

{ update statement : S2
  initialization statement : S1
  state δ
  loop body : S3
  condition : E

5) C switch

[[switch (E) { case C1; S1; case C2; S2; ⋯ default : SD ; }]] (6) =

let v = [[E]] (6) in

case v of

C1 : [[S1]] (6)

C2 : [[S2]] (6)

...

default : [[SD]] (6)

- expression : E
- state : 6
- value : v
- case labels : C1, C2, ⋯
- corresponding statement : S1, S2, ⋯
- default case statement : SD
- state : 6

23. weakest precondition for each condition

1) WP → $2 \times (b-1) - 1 > 0$

$2b - 3 > 0$

$b > \frac{3}{2}$

2) WP → $\frac{c+10}{3} > 6$

$c + 10 > 18$

$c > 8$

3) WP → $a + 2 \times b - 1 > 1$

$a > 2 - 2b$

4) WP → $2xy + x - 1 > 11$

$x > 12 - 2y$